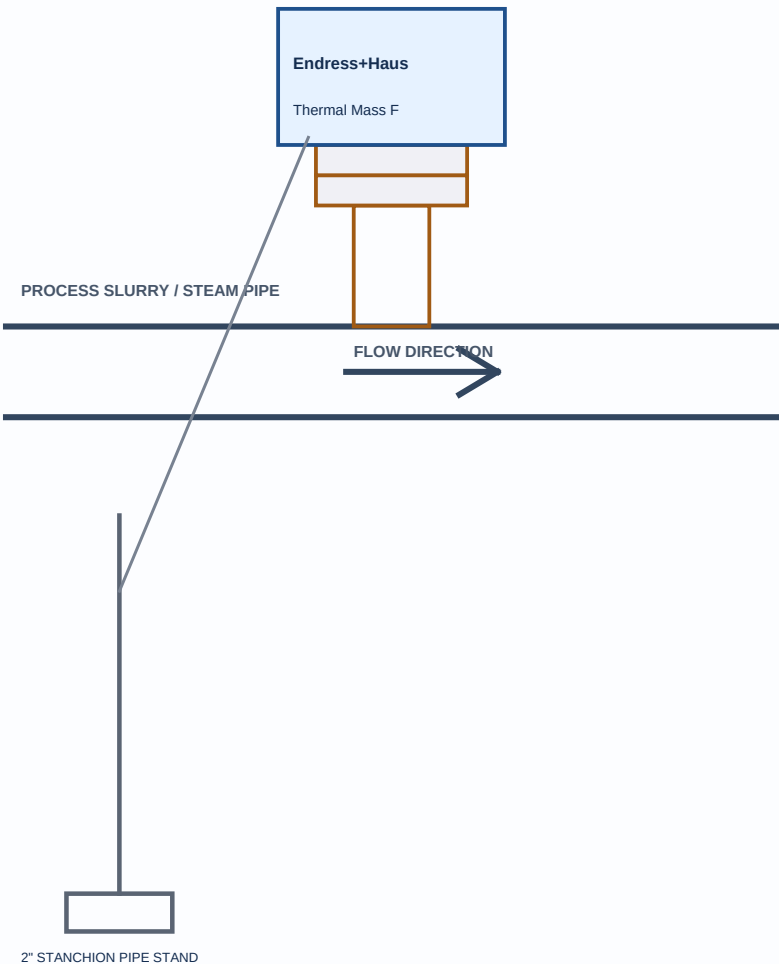


TEMPLATE TYP-09: THERMAL MASS FLOWMETER

TYPICAL TAGS: FT-80301 Series

SERVICE: LPG / Natural Gas Boiler Fuel Consumption | PLC I/O: 1756-IF16 (AI Gas Flow) & 1756-IB32 (DI Alarm)

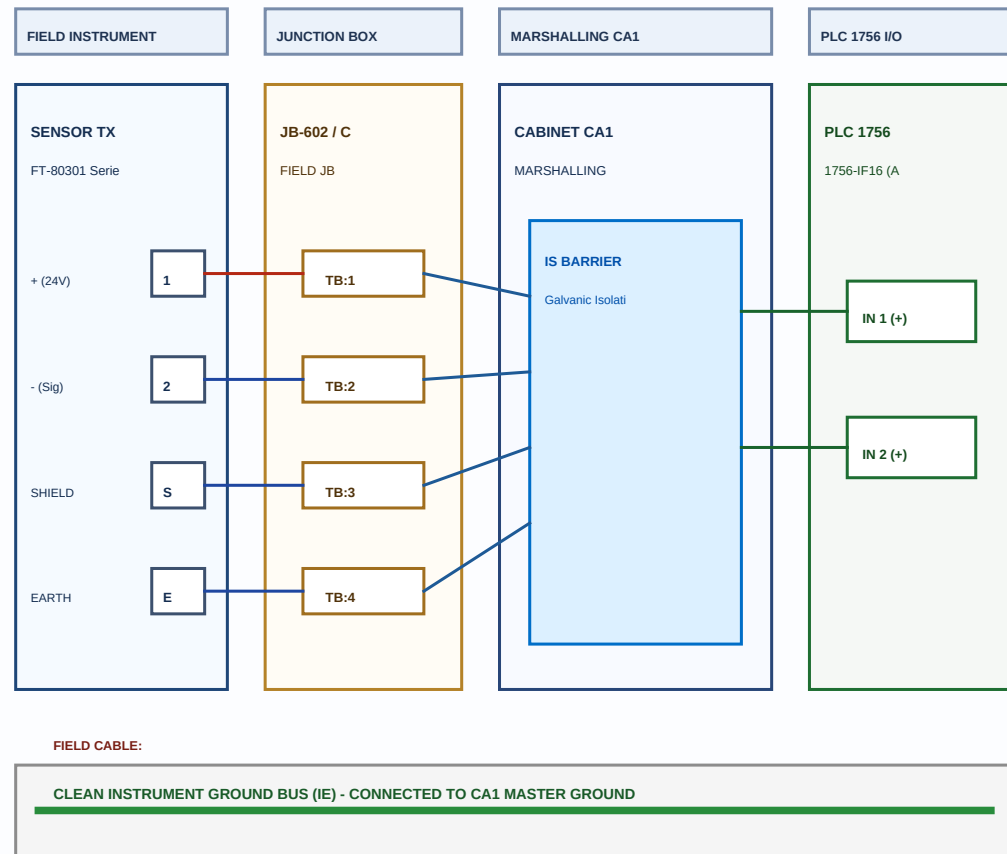
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN100 Flange / 1" NPT Retract..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP67 Aluminum Encl..
Hazardous Class:	ATEX II 2G Ex db eb ia IIC T..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART Standard Nm³/h Flow
Power Supply:	24V DC / 12W (4-Wire Electronics)
Field Cable Type:	2Px1.5mm² Shielded Armored (Belden 8777)
Destination JB:	JB-602 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	Galvanic Isolation / Ex d Installation
PLC Module:	1756-IF16 (AI Gas Flow) & 1756-IB32 (D..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: t-mass F 300 / I 300 (6F3B / 6I3B Insertion)

SERVICE: LPG / Natural Gas Boiler Fuel Consumption

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	t-mass 300 Thermal Flowmeter T E+H Insertion Probe w/ Hot-Tap		1 EA
2	Weld-in Boss 1" NPT Female Thr ASTM A105 / Carbon Steel Boss		1 EA
3	Isolation Ball Valve 1" Full B SS316 Cl.300 Gas Ball Valve		1 EA
4	Cable Gland M20x1.5 Ex d Flame Brass Gland for Armored Cable		2 EA
5	Grounding Cable 6mm² w/ Lug Copper Earthing Conductor		1 EA

ENGINEERING INSTALLATION NOTES

- Install probe at centerline of pipe (0.5 x Inside Diameter) matching flow direction arrow exactly.
- Verify straight pipe run: min. 20D upstream and 5D downstream of insertion tap.
- Equipotential bonding must be connected between meter body and gas piping system.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTTHERMAL MASS GAS FLOWMETER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONDWG NO: KAL-JC-TYP-009
REV: 3.6 DATE: 2026-09-08PROJECT NO:
R5THS00138-JCSHEET NO:
11 / 25